F01 Numbers, names and predicting Python output

Week 1, fourth teaching day. Complete ML03 first.

Today's question. Can I predict every line Python prints before I run it?

> **WHERE THIS IS USED**
> A good programmer reads code and predicts its output before pressing run. A mobile money app keeps an amount, a name and a date in named boxes, and each one must be right. Can you predict every line before it runs?

Words for today

Word	What it means	Think of it like this
scalar	One number.	One price on a label.
name	A label that points to a value.	A tag on a box.
top to bottom	Python reads one line after another and does not look ahead.	Reading a recipe in order.
update	Giving an existing name a new value.	Crossing out a price and writing a new one.
string	Text inside quotation marks.	A word on a label.
prediction on paper	Writing what you expect before you run.	Guessing the match score before kick-off.

Learn the idea

Python reads instructions from the top to the bottom, one at a time. A name points to a value. Changing a name does not change a result that was already calculated. Multiplication uses * and division uses /. The text in quotation marks is a string even when it looks like a number. print shows its input.

Start on paper

Draw three pens and label each UGX 1,000. Add them by hand: 1,000 + 1,000 + 1,000 is 3,000. Explain why 3 * 1000 gives the same total. Then write on paper every line the code below will print, before you run it.

Worked example

```
units = 3
price = 1000
total = units * price
print(total)
print(total / units)
print("1000")
units = 4
print(total)
total = units * price
print(total)
```

Run it, then compare: 3000, 1000.0, 1000, 3000 and 4000, on separate lines. The quoted 1000 is text. Changing units alone does not recalculate an earlier total.

Your turn

In a new code cell calculate the price of 5 books at UGX 3000 each and print it. Change the quantity to 2 and recalculate. In a text cell explain why the old total did not change until you recalculated it. Save F01.ipynb.

Run all cells from the top, save, close and reopen. Show the instructor your output and explanation.